

Mandar Ambulkar

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EDUCATION

Northeastern University, San Jose, CA - Master of Science in Computer Science, GPA: 3.87/4.0

Jan 2024 – Dec 2025

Relevant Courses: Pattern Recognition, Computer Vision, NLP, Data Mining, Algorithms, Programming Design Paradigm.

Vishwakarma Institute of Technology, Pune, India

Aug 2019 – May 2023

Bachelor of Technology, Information Technology, GPA: 8.47/10.0

Relevant Coursework: Database Management Systems, Machine Learning, Deep Learning, Networking, Web Development, Cloud Computing, Image Processing, Network Security, Computer Architecture and Operating System Data Communication and Networking.

EXPERIENCE

Software Development Engineer, PASAL Inc. | Remote

May 2026 – Present

- **Architected and deployed** a scalable, distributed, cloud-native AI outbound email pipeline using Next.js 15, Inngest durable functions, and LLMs (Gemini & GPT-4o-mini), automating the delivery of ~500 highly personalized emails/month while maintaining an optimized operating budget of ~\$85/month.
- **Engineered a robust database layer** leveraging Neon Postgres and Drizzle ORM, integrating a non-bypassable pre-send compliance verification framework to ensure 100% adherence to CAN-SPAM and AI-content regulations.

Software Development Engineer, SurfSense (NU ChatBot Project) | San Jose

September 2025 – December 2025

- **Productionized an open-source RAG platform** with RBAC, domain-restricted authentication, and privacy-aware retrieval controls on Azure (App Services, PostgreSQL, AD), supporting **secure, compliant multi-tenant access** for over **1,000 institutional users**.
- **Reduced end-to-end query latency** from 320 ms to <180 ms by optimizing a hybrid RAG pipeline using pgvector, rerankers, intelligent response caching, and asynchronous FastAPI endpoints **under concurrent production workloads**.
- **Improved frontend performance by 40%** by designing a **modular** React + Zustand architecture with multi-session chat memory visualization for scalable context handling across thousands of active interactions.

Graduate Teaching Assistant, Khoury College of Computer Science | Northeastern University September 2024 – April 2025

- Led a team of 3 TAs to support 40+ students in algorithm design and analysis (CS5800/CS5008) through lab sessions, office hours, and programming guidance while collaborating with faculty on course materials.

Software Developer, Reticulo Sports Technology Pvt. Ltd. | Pune, India

January 2023- December 2023

- **Improved product categorization efficiency by 25%** by classifying 12,000+ images into 12 categories with MobileNetV3, cutting manual verification time by 30%.
- **Boosted GPU inference speed by 40%** using Nvidia Triton and CUDA, reducing latency from 15ms to 10ms while enhancing throughput by 20%.
- **Deployed optimized deep learning models** on low-power mobile devices, reducing memory usage by 50% and halving model size (1.2GB → 600MB).
- **Enhanced app scalability** to support 50,000+ daily interactions with minimal latency while raising accuracy by 10% and reducing training time by 20% through dataset refinement and fine-tuning.

PROJECTS

Algorithm RAG System [\[Link\]](#)

San Jose | April 2025

Retrieval Augmented Generation system for algorithm recommendations

- Achieved 95% recommendation accuracy by implementing semantic search with FAISS vector storage, processing 500+ algorithm chunks with sub-100ms query latency.
- **Reduced search complexity by 40%** and accelerated workflows by 50% by building a hybrid retrieval FastAPI service combining vector similarity and keyword matching across 10 categories for 100+ concurrent users.

AI Demo Builder (Cloud Computing Project) [\[Link\]](#)

San Jose | December 2025

Cloud-native, serverless AI platform for automated demo generation from GitHub repositories

- Architected a distributed serverless platform using 18 event-driven AWS Lambda microservices with S3 and DynamoDB, provisioned end-to-end via AWS CDK for reproducible, production-grade infrastructure.
- **Enhanced system robustness** by implementing **validation, failure isolation, and cleanup services**, successfully delivering a fully integrated system verified through rigorous **end-to-end testing**.

ACHIEVEMENTS / PUBLICATIONS

- **3rd place, Northeastern Student Research Showcase, 2024: Multi Modal Hand Gesture Recognition** [\[link\]](#)
 - Achieved 97.25% validation accuracy by developing ensemble of 34 binary classifiers with priority-based fusion, processing 119GB dataset with 25-30 FPS real-time performance and <40ms latency.
 - Reduced computational overhead by 60% through implementing PCA-based feature extraction pipeline, maintaining 2GB memory footprint while processing 1M+ multi-modal images across 35 gesture classes.
- **Publication: Number Plate Recognition System, IC-ICN 2022** [\[link\]](#)
- **Publication: KhetiKaksha (Farmer Hub), IC-ICN 2022** [\[link\]](#)

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript

Frameworks & Libraries: Keras, TensorFlow, PyTorch, Scikit-learn, OpenVINO, Torch, React, Next.js, FastAPI, Flask, Node.js, Tailwind CSS, LangChain, Pandas, NumPy, BeautifulSoup, Uvicorn, Gunicorn, Jest, ESLint

AI/ML & NLP: LLM Integration, RAG Systems, Vector Embeddings, FAISS, Semantic Search, NLP Processing, Prompt Engineering

Databases: MySQL, PostgreSQL, SQLite, MongoDB, Supabase, Vector Databases

Cloud & DevOps: AWS, Google Cloud, Docker, Kubernetes, OpenShift, Terraform, Ansible, Helm, CI/CD (Jenkins, GitLab CI), GitOps, Vercel

Performance & CI/CD: Nvidia Triton, TensorRT, Git, GitLab, Jenkins, Prometheus, Grafana

Web Technologies: REST APIs, OAuth Integration, WebSockets, State Management, API Integration

Distributed Systems & Networking: Microservices Architecture, Distributed Systems, Low-latency Systems, Edge Computing (foundational)